Data Structures & Algorithms (CST-003) - Unit 1

Unit 1 - Introduction, Analysis & Searching - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: what data structures are, how to measure an algorithm (complexity), and the two basic searches.

Key ideas: A **data structure** organises data so operations stay fast; an **ADT** (abstract data type) defines *what* operations exist, not *how* they are coded. **Time complexity** counts how work grows with input size n: O(1) constant, O(log n), O(n), O(n log n), O(n²) - memorise this order. **Big-O** = upper bound (worst case), **Omega** = lower bound, **Theta** = tight bound. **Linear search** checks each element: O(n). **Binary search** needs a sorted array, halves the range each step: O(log n) - learn the mid = (low + high)/2 loop and its dry runs.

Exam pointers: expect "find the complexity of this loop" questions (nested loops on n give O(n²), halving loops give O(log n)) and a binary-search dry run on a given sorted array. Writing linear vs binary search with their conditions is a classic 5-marker.

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